

# An Infinite Pell Family for $y^2 + x^2y + xz^2 - 3 = 0$

## Abstract

We give a completely explicit infinite family of integer solutions to

$$y^2 + x^2y + xz^2 - 3 = 0.$$

The construction is elementary: fixing  $x = -2$  reduces the equation to the generalized Pell equation  $(y + 2)^2 - 2z^2 = 7$ . A single solution of the latter is propagated by an explicit norm-preserving recurrence. No general theorem about Pell equations is required.

## 1 The infinite family

**Lemma 1** (Norm-preserving recurrence). *Let  $(U_n, V_n)_{n \geq 0}$  be defined by*

$$(U_0, V_0) = (3, 1), \quad \begin{cases} U_{n+1} = 3U_n + 4V_n, \\ V_{n+1} = 2U_n + 3V_n. \end{cases}$$

*Then, for every  $n \geq 0$ ,*

$$U_n^2 - 2V_n^2 = 7.$$

*Moreover,  $U_n$  and  $V_n$  are positive, and  $(V_n)_{n \geq 0}$  is strictly increasing.*

*Proof.* For arbitrary integers  $U, V$  one has the identity

$$\begin{aligned} (3U + 4V)^2 - 2(2U + 3V)^2 &= 9U^2 + 24UV + 16V^2 \\ &\quad - 2(4U^2 + 12UV + 9V^2) \\ &= U^2 - 2V^2. \end{aligned}$$

Thus the recurrence preserves the quadratic form  $U^2 - 2V^2$ . Since  $U_0^2 - 2V_0^2 = 9 - 2 = 7$ , induction gives  $U_n^2 - 2V_n^2 = 7$  for all  $n$ .

The recurrence has integral coefficients and integral initial values, so  $U_n, V_n \in \mathbb{Z}$  for all  $n$ . The initial values are positive, and the recurrence has positive coefficients, so  $U_n, V_n > 0$  for all  $n$ . Finally,

$$V_{n+1} - V_n = 2U_n + 2V_n > 0,$$

so  $V_{n+1} > V_n$  for every  $n$ . □

**Theorem 1.** *For every integer  $n \geq 0$ , the two triples*

$$(x, y, z) = (-2, U_n - 2, \pm V_n)$$

*are integer solutions of*

$$y^2 + x^2y + xz^2 - 3 = 0.$$

*Consequently, the equation has infinitely many integer solutions.*

*Proof.* Fix  $n \geq 0$  and put  $x = -2$ ,  $y = U_n - 2$ , and  $z = \pm V_n$ . By Lemma 1,

$$\begin{aligned} y^2 + x^2y + xz^2 - 3 &= (U_n - 2)^2 + 4(U_n - 2) - 2V_n^2 - 3 \\ &= U_n^2 - 2V_n^2 - 7 \\ &= 0. \end{aligned}$$

Hence every displayed triple is a solution. The positive integers  $V_n$  are strictly increasing by Lemma 1; therefore the triples with  $z = V_n$  are pairwise distinct. This proves infinitude. □

The first values are

$$(U_n, V_n) = (3, 1), (13, 9), (75, 53), (437, 309), \dots,$$

which yield

$$(-2, 1, \pm 1), \quad (-2, 11, \pm 9), \quad (-2, 73, \pm 53), \quad (-2, 435, \pm 309), \dots$$

## 2 How the construction was found

The original expression is quadratic in  $y$ . Completing the square is therefore the natural first move:

$$4(y^2 + x^2y + xz^2 - 3) = 0 \iff (2y + x^2)^2 + 4xz^2 = x^4 + 12. \quad (1)$$

The sign of  $x$  now becomes decisive. When  $x > 0$ , the two square terms on the left of (1) have the same sign, so this specialization does not naturally produce an unbounded indefinite quadratic equation. To obtain a Pell-type mechanism, one instead takes  $x < 0$ . Writing  $x = -m$  with  $m > 0$  gives

$$(2y + m^2)^2 - 4mz^2 = m^4 + 12. \quad (2)$$

The smallest choices illustrate what is needed. For  $m = 1$ , equation (2) becomes

$$(2y + 1)^2 - (2z)^2 = 13,$$

a difference of two integral squares. It factors as  $(2y + 1 - 2z)(2y + 1 + 2z) = 13$ , so this route can give only finitely many pairs.

The next choice,  $m = 2$ , is qualitatively different. Equation (2) becomes

$$(2y + 4)^2 - 8z^2 = 28,$$

and division by 4 gives

$$(y + 2)^2 - 2z^2 = 7. \quad (3)$$

This is an indefinite quadratic equation with the obvious seed

$$3^2 - 2 \cdot 1^2 = 7.$$

What remains is to reproduce this seed indefinitely without changing the value 7.

For this purpose, write a pair  $(U, V)$  formally as  $U + V\sqrt{2}$  and define its norm by

$$N(U + V\sqrt{2}) = U^2 - 2V^2.$$

Indeed, since

$$(U + V\sqrt{2})(A + B\sqrt{2}) = (UA + 2VB) + (UB + VA)\sqrt{2},$$

one obtains by direct expansion

$$\begin{aligned} N((U + V\sqrt{2})(A + B\sqrt{2})) &= (UA + 2VB)^2 - 2(UB + VA)^2 \\ &= (U^2 - 2V^2)(A^2 - 2B^2) \\ &= N(U + V\sqrt{2})N(A + B\sqrt{2}). \end{aligned}$$

Hence multiplication by any element of norm 1 preserves equation (3). The small identity

$$3^2 - 2 \cdot 2^2 = 1$$

provides such an element, namely  $3 + 2\sqrt{2}$ . Multiplying gives

$$(U + V\sqrt{2})(3 + 2\sqrt{2}) = (3U + 4V) + (2U + 3V)\sqrt{2},$$

which is exactly the recurrence in Lemma 1. Starting from  $3 + \sqrt{2}$ , whose norm is 7, repeated multiplication by  $3 + 2\sqrt{2}$  produces

$$U_n + V_n\sqrt{2} = (3 + \sqrt{2})(3 + 2\sqrt{2})^n.$$

The recurrence proof in Section 1 then establishes integrality, preservation of the norm, and strict growth without appealing to any external result on Pell equations.